FINITE DISCRETE SCHEMES ARE AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2308-2315, following the statement at 2301-2306.

Editorial note. The step 'By lemma-basis-affine-opens it is affine' is read as: the basis supplies an affine open W with x_i in W subset {x_i}, forcing W = {x_i}; mechanical reading of the basis statement, flagged here.

Allowed standard prerequisites (editorial).

- Affine opens form a basis. lemma-basis-affine-opens (same chapter, cited at ~2311). Each singleton {x_i} is affine.
- A disjoint union of two affines is affine. lemma-disjoint-union-affines (card L26402, cited at ~2314). Concluding engine, iterated.

Source reference. Original lemma 01IT: sources/stacks-schemes/pinned-original-source.tex, source lines 2169–2169. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I5: sources/stacks-schemes/pinned-original-source.tex, source lines 1246–1246. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme whose underlying topological space is a finite discrete set. Then X is affine.*

Proof. Say $X = \{x_1, \ldots, x_n\}$. Then $U_i = \{x_i\}$ is an open neighbourhood of x_i. By Lemma 01IT it is affine. Hence X is a finite disjoint union of affine schemes, and hence is affine by Lemma 01I5. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.